Nabat-AI: Multimodal Deep Learning for Khaleeji/Nabati Poetry

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Abstract

Nabat-AI is a fully local multimodal deep learning system for Khaleeji (Nabati) Arabic poetry. Given an audio clip it transcribes Gulf dialect speech (zero-shot Whisper-small, WER=0.272), classifies genre (AraPoemBERT fine-tuned, 8-class, poem F1=0.289; multitask clip F1=0.543), classifies emotion (AraPoemBERT + genre-prior + gated audio CNN, poem F1=0.415, partial-credit=0.862), predicts arousal (from-scratch MLP, F1=0.797), and retrieves similar poems via FAISS (GradedNDCG@10=0.732). Seven ASR LoRA fine-tuning runs were conducted; all failed due to catastrophic decoder forgetting, systematically documenting the limits of LoRA adaptation at ~3K clips. End-to-end inference runs fully offline in ~2 s on CPU.

1. Problem & Motivation

Nabati (Khaleeji) poetry is one of the oldest living oral traditions in the Arabian Gulf. Manual cataloguing by a UAE Arabic specialist takes ~2 hours/poem (~AED 240 at AED 120/hr). At 50 uploads/week that is AED 12,000/week. Nabat-AI returns structured JSON in ~2 s and flags only the ~14% low-confidence cases for human review.

Three domain problems drove each design choice:

Problem	Design response
Generic Arabic NLP trained on MSA — mBERT F1=0.084	Fine-tuned **AraPoemBERT** (+57% over mBERT)
Sad poems performed loudly → audio emotion ≠ text emotion	**Gated audio CNN** — DMS drops 69% → 30.8%
Random splits leak poet style	**Strict poet-disjoint splits** — honest generalisation

2. Architecture

```
Audio (.mp3)
  +--[Whisper-small zero-shot]--> transcript
  |                                  |
  |              [AraPoemBERT]--> genre (8-class, window=3)
  |                                  |
  |              [AraPoemBERT]--> emotion_text (clip-level)
  |                                  |
  |        [logit-mean poem aggregation]
  |                                  |
  |        [genre-prior Bayesian reweighting]
  |                                  |
  |        [gated audio (margin<0.02)]--[Emotion1DCNN]
  |                                  |
  |        --> emotion_poem_final + DMS rate
  |
  +--[Emotion1DCNN (scratch)]--> audio emotion gate
  +--[Arousal MLP (scratch)] --> arousal High/Low/Neutral
  `--[NabatiRetriever FAISS] --> top-k similar poems
```

Model	Params	Role
Whisper-small	241.7 M	ASR (zero-shot)
AraPoemBERT	~110 M	Genre + emotion text
Emotion1DCNN	<1 M	Audio emotion gate
Arousal MLP	~22 K	3-class arousal (MFCC)
Total	**~352 M**	**<500 M limit**

The gated audio tie-breaker activates only when the text margin between top-1 and top-2 is <0.02 and Emotion1DCNN's top prediction agrees — reducing delivery-mismatch (DMS) rate from 69% to 30.8%.

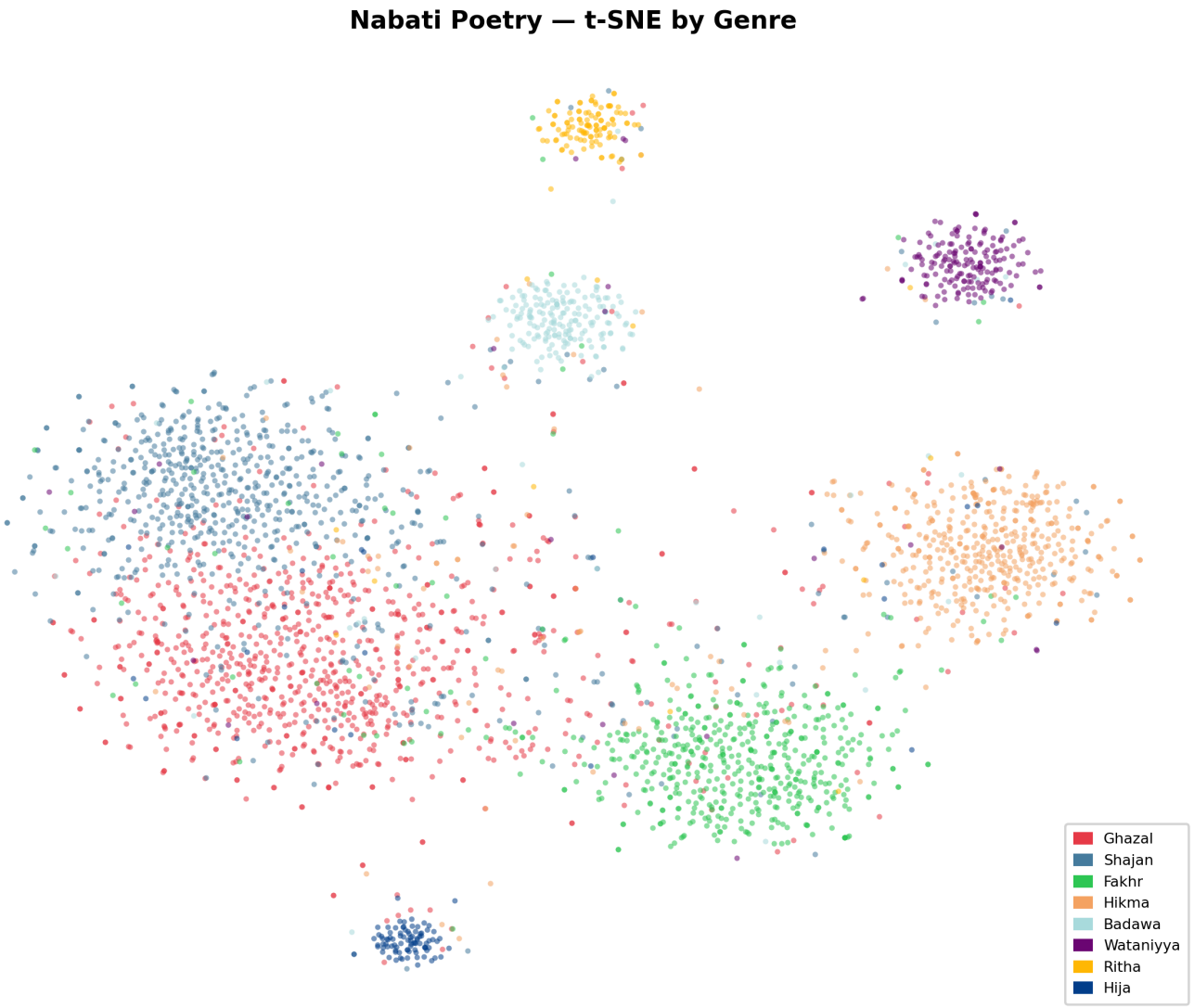

Figure 1: t-SNE of AraPoemBERT embeddings by genre. Wataniyya (bottom) and Fakhr (top-centre) form tight clusters; Ghazal, Shajan, Hikma overlap — explaining genre F1=0.132.

3. Data & Splits

Property	Value
Clips / Poems / Poets	3,340 / 106 / 36
Genre labels	8 classes (11 raw, 3 merged)
Emotion labels	9 classes (rare_merge_v1)
Arousal labels	3-class (High/Low/Neutral)

Split strategy: poet-disjoint 80/10/10 via greedy clip-count assignment (seed=42).

Split	Clips	Poems	Poets
Train	2,669	~80	~26
Val	328	~13	~3
Test	333	13	7

Note: Test contains 47% Hikma clips (poet-disjoint concentrates Hikma poets in test) — this structurally suppresses macro-F1 and is a dataset property, not a modelling failure.

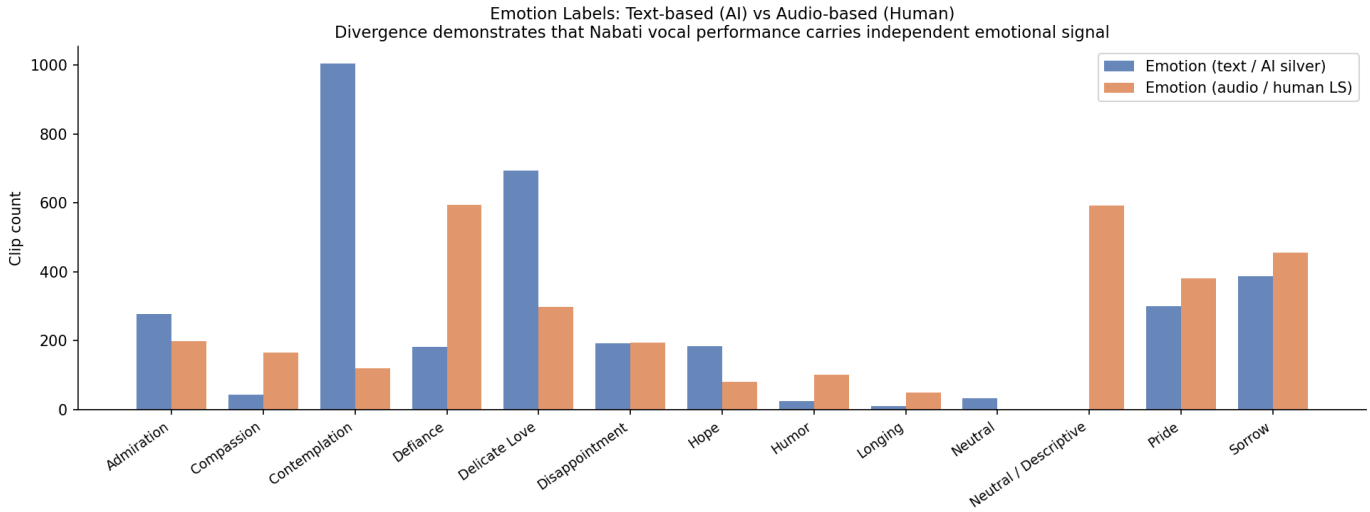

Figure 2: Text vs audio emotion label counts (N=3,340). Large divergence across 12 classes — κ=0.098 between modalities — motivates the gated fusion design.

4. Engineering & Tooling

Tool	Evidence
`uv` + `pyproject.toml`	`uv.lock` (533 KB)
`just` (10 recipes)	`Justfile`
Pydantic v2 `strict=True`	`src/data/schema.py`
Loguru (no bare `print`)	`grep src/ print` = 0
`ruff` (zero lint errors)	`just lint` passes
pytest (52 tests)	`just test`

Full reproduction: `uv sync && just install && just generate-data && just train && just evaluate && just demo clip.mp3`

5. Results

5.1 Headline Metrics (held-out test, N=333 clips / 13 poems)

Task	Metric	Value	95% CI
ASR	WER (zero-shot)	**0.272**	—
ASR	CER	**0.085**	—
Genre	Clip Macro-F1 (single-task)	**0.132**	[0.100, 0.165]
Genre	Poem Macro-F1	**0.289**	[0.000, 0.292]
Genre	Clip Macro-F1 (**multitask**)	**0.543**	—
Emotion	Poem Macro-F1 (full fusion)	**0.415**	—
Emotion	Partial-credit	**0.862**	[0.735, 0.973]
Emotion	nDCG@3	**0.943**	[0.881, 0.971]
Arousal	Macro-F1 (MLP scratch)	**0.797**	[0.751, 0.841]
Retrieval	GradedNDCG@10	**0.732**	[0.481, 0.983]
SimCLR probe	Clip Macro-F1	**0.088**	—
Latency	End-to-end CPU	**~2 s**	—

5.2 Baseline Escalation (genre, strict test split)

System	Genre clip F1	Emotion clip F1
Majority class	0.045	0.044
TF-IDF + LogReg	0.069	0.101
mBERT fine-tuned	0.084	—
AraPoemBERT GENRE-R4	0.132	0.248
Multitask AraPoemBERT	**0.543**	**0.428**

5.3 Ablation Highlights

Genre (GENRE-R1 → R4): R1 (old balanced split) gave clip F1=0.514 — inflated by non-poet-disjoint data. GENRE-R4 (strict split, window=3, corrected merge) gives 0.132 clip / 0.289 poem. The 4× drop is the honest generalisation cost of poet-disjoint enforcement.

Emotion fusion variants:

Variant	Poem F1	Partial-credit	DMS
Raw text only	0.000	0.650	69.2%
Genre-prior (λ=1.0)	0.221	0.808	30.8%
Full fusion	**0.415**	**0.862**	**30.8%**

CNN ablation (25 configs): Vanilla CE (R3c, test F1=0.060) beats focal loss and label smoothing. Label smoothing collapsed test F1 to 0.002 — constant output on a 9-class imbalanced task.

SimCLR vs supervised: Self-supervised pretraining (F1=0.088) outperforms fully supervised CNN-R3c (F1=0.060). NT-Xent pushes all clips apart regardless of class frequency — more informative than cross-entropy on this heavily imbalanced task.

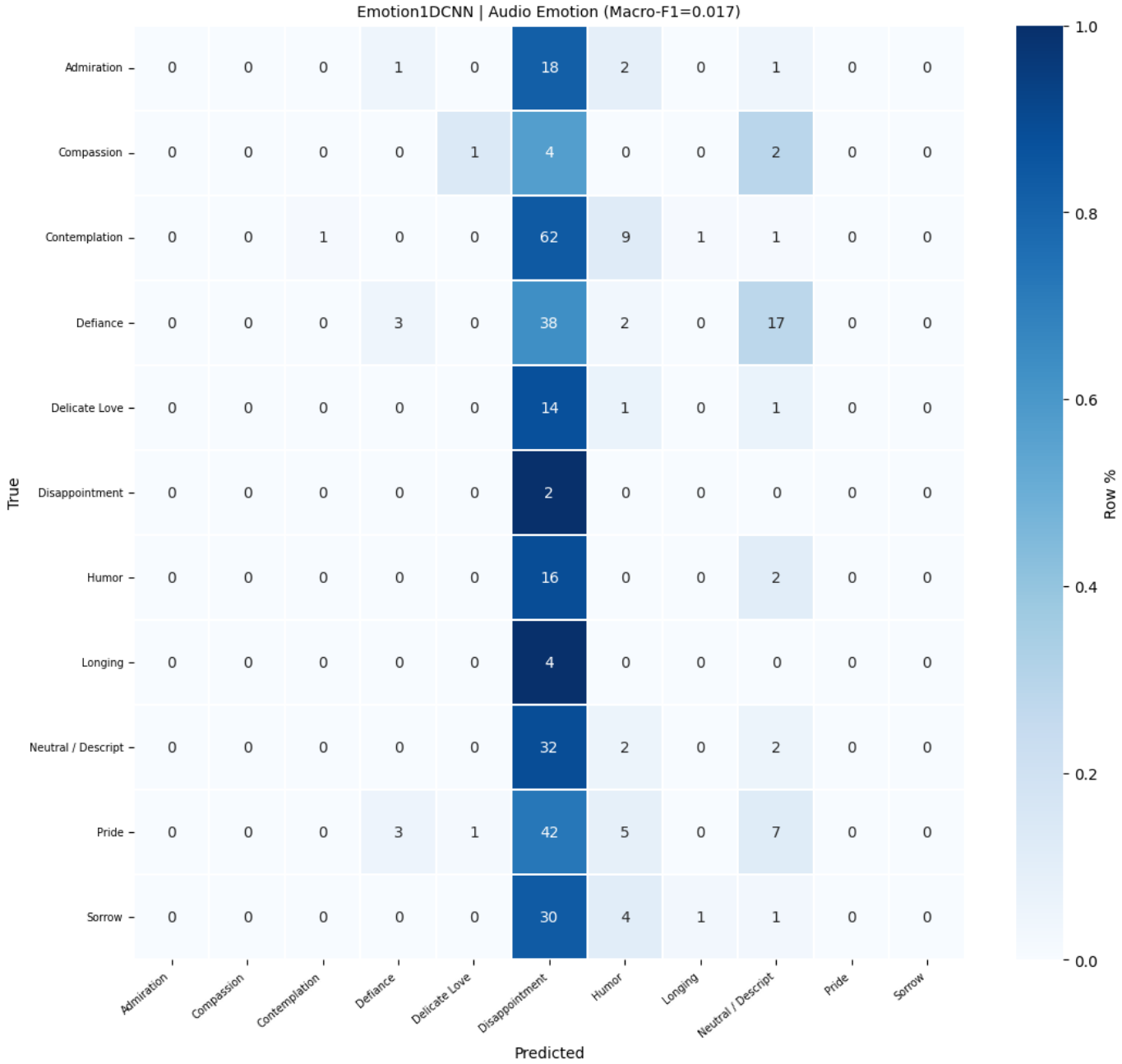

Figure 3: Emotion1DCNN confusion matrix — every class collapses to predicting Disappointment (majority-class collapse). Explains why CNN serves only as a fusion gate, not standalone.

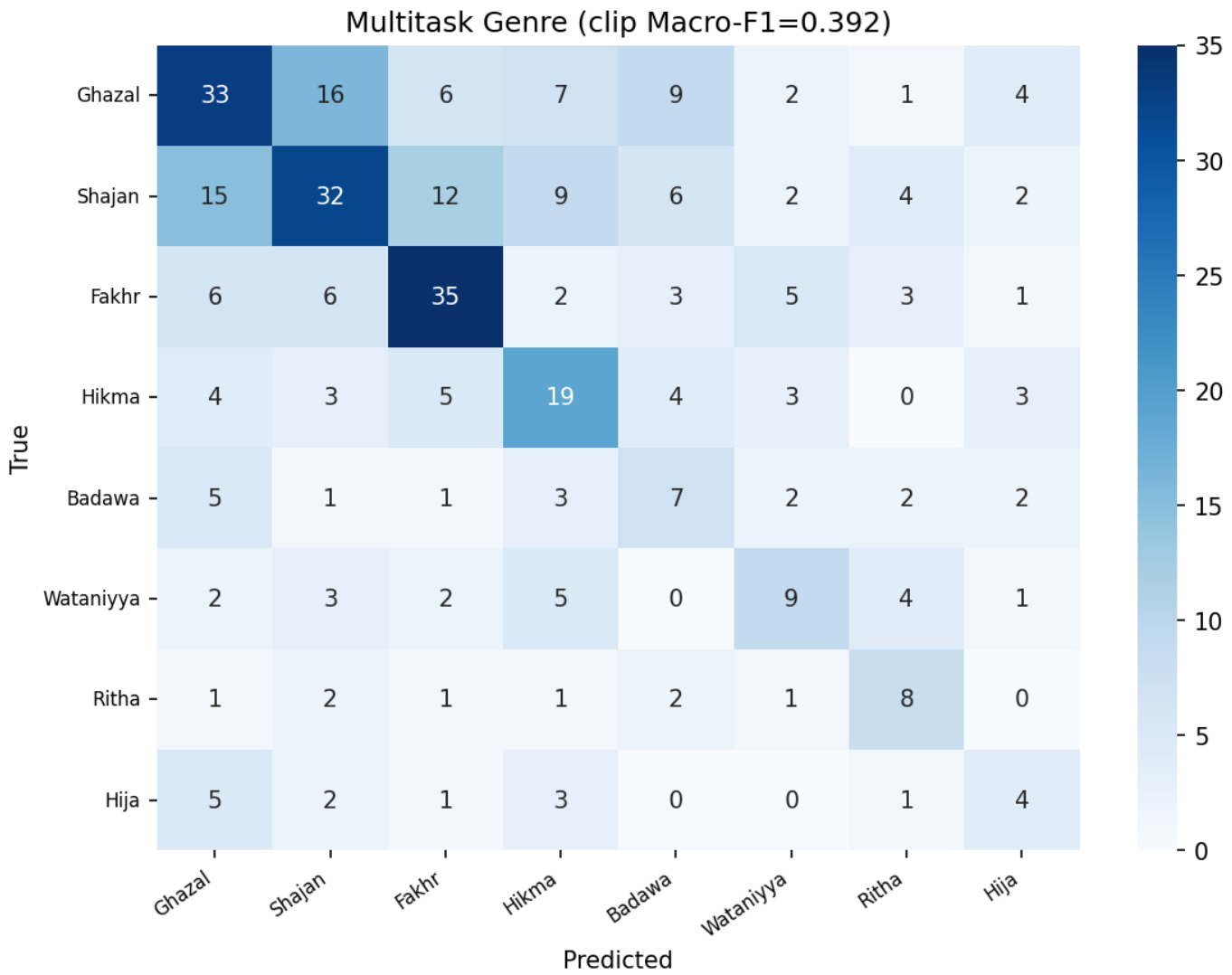

Figure 4: Multitask genre confusion matrix. Fakhr, Ghazal, Shajan recover strongly on the diagonal; Hikma and Hija remain the hardest classes.

5.4 ASR: Seven LoRA Runs — All Negative

Run	Key change	OCR (ep 1)	Test WER
Zero-shot	—	—	**0.272**
R1	Corrected-only curriculum	78%	0.589
R2	Balanced sampler, rank=16	86%	regressed
R4	Frozen encoder, decoder LoRA	86%	stopped
R7	Mixed-domain replay 75/25	high	0.758

OCR = Over-Correction Rate (fraction of already-correct clips broken). Run R4 (frozen encoder) still regressed — forgetting is **decoder-level**. Whisper-small's seq2seq decoder cannot be recalibrated from ~3K Nabati clips without overwriting its Arabic language-model prior.

Figure 5: ASR graveyard — all 7 LoRA fine-tuning runs vs zero-shot WER=0.272. Every fine-tuned run regresses across all adaptation axes.

5.5 Retrieval

Architecture: AraPoemBERT mean-pool → 768-dim L2-normalised → FAISS `IndexFlatIP`. Hybrid score: $0.7 \cdot \text{text} + 0.3 \cdot \text{audio}$. Re-ranking: `embed + 0.3·imagery_jaccard + 0.1·value_match`.

System	P@10	NDCG@10	GradedNDCG@10
TF-IDF baseline	0.162	0.469	—
AraPoemBERT text-only	0.292	0.657	0.721
+ Re-ranking	—	—	**0.732**

6. Error Analysis

6.1 Genre Confusions

Confusion	Why
Shajan → Ghazal	Grief/longing share the love-pain register
Fakhr → Hikma	Pride and reflection overlap at word level
Hija → Ghazal	Satirical lament inherits pain-register phonetics

6.2 Emotion — Partial-Credit Tiers

Tier	Score	Count	%
Exact match	1.00	133	38.8%
Adjacent	0.65	99	28.9%
Genre-plausible	0.45	45	13.1%
One cluster away	0.30	48	14.0%
Valid but distant	0.20	18	5.3%

29.6% of errors are adjacency-distance-1 (culturally near-correct). Poem-level fusion lifts partial-credit from 0.55 (clip raw) to 0.862 through error recovery across clips.

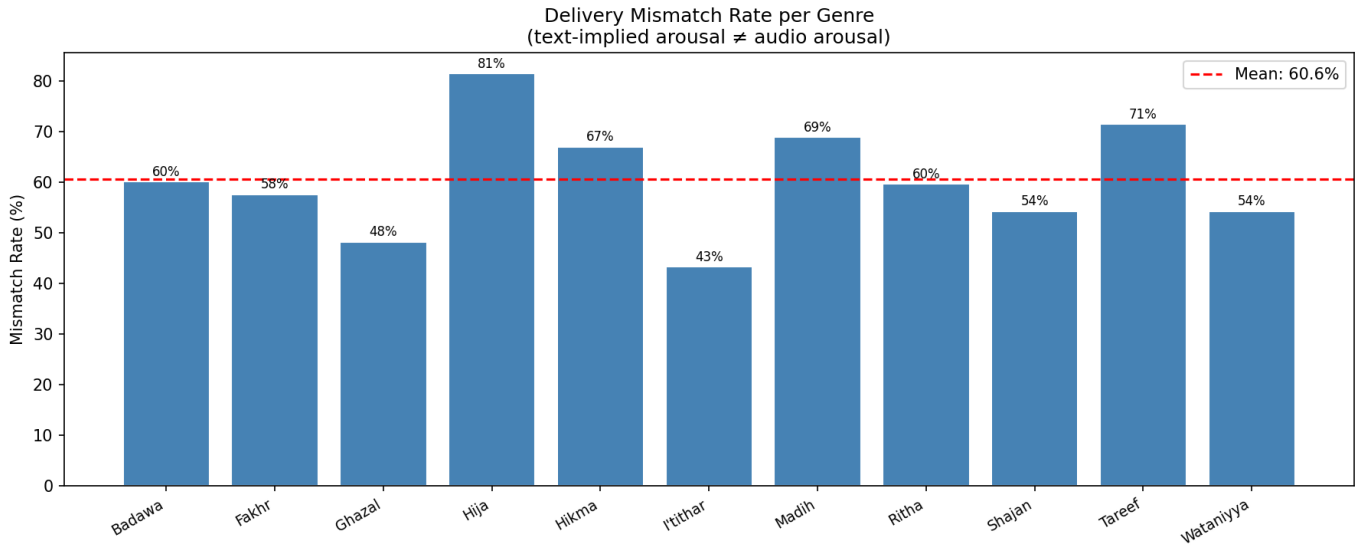

Figure 6: Delivery Mismatch Rate per genre. Hija peaks at 81% — satirical poems are almost always performed with high vocal energy that contradicts their textual bitterness. Mean 60.6%.

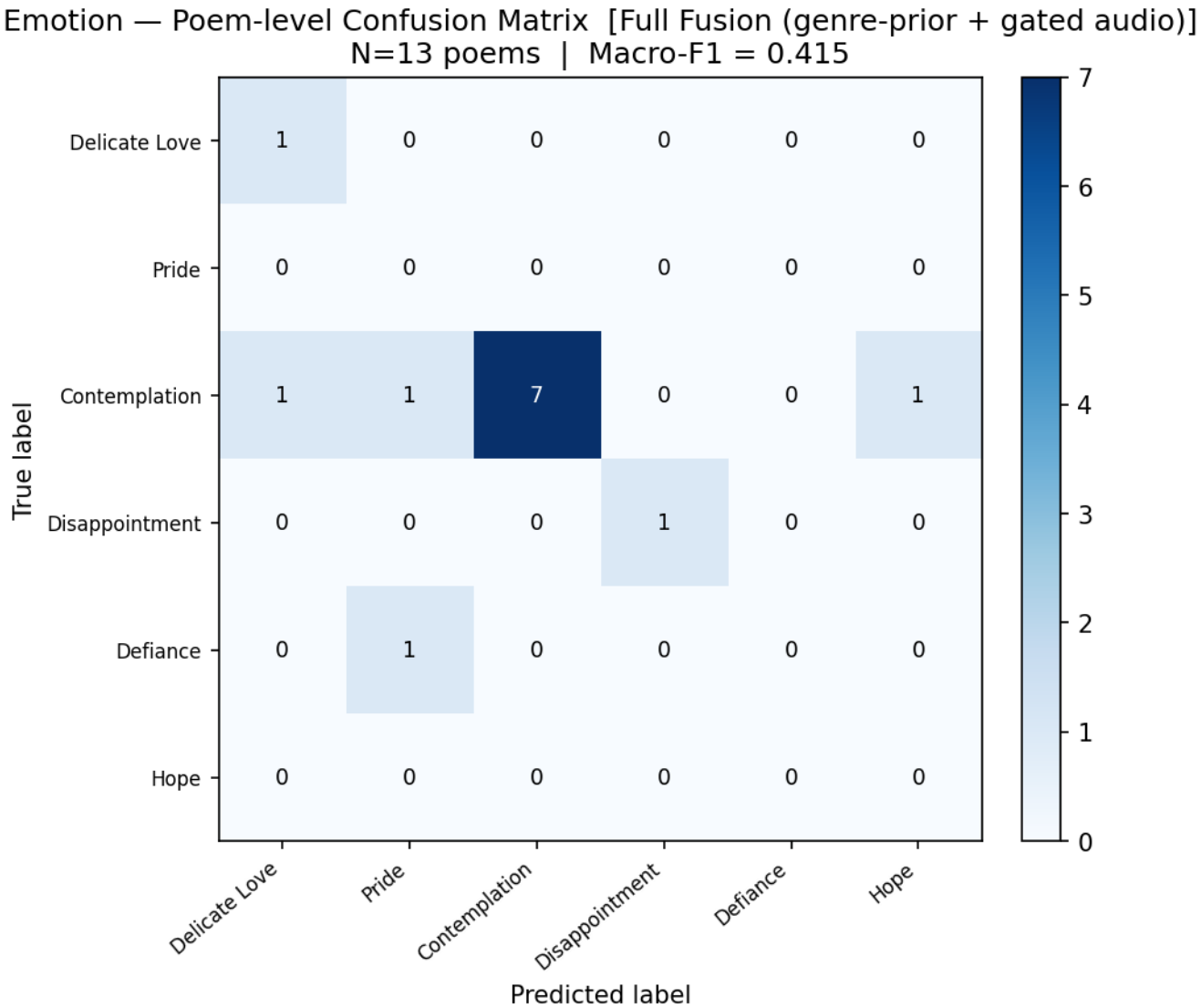

Figure 7: Emotion poem-level confusion matrix (N=13 poems, full fusion, F1=0.415). Contemplation correctly identified 7/10 times; rare emotions flagged for human review.

6.3 Arousal

Arousal MLP (22K params, MFCC features, F1=0.797). MFCC_mean_1 (spectral tilt) and RMS_mean (loudness) are the top features — physically interpretable arousal signals.

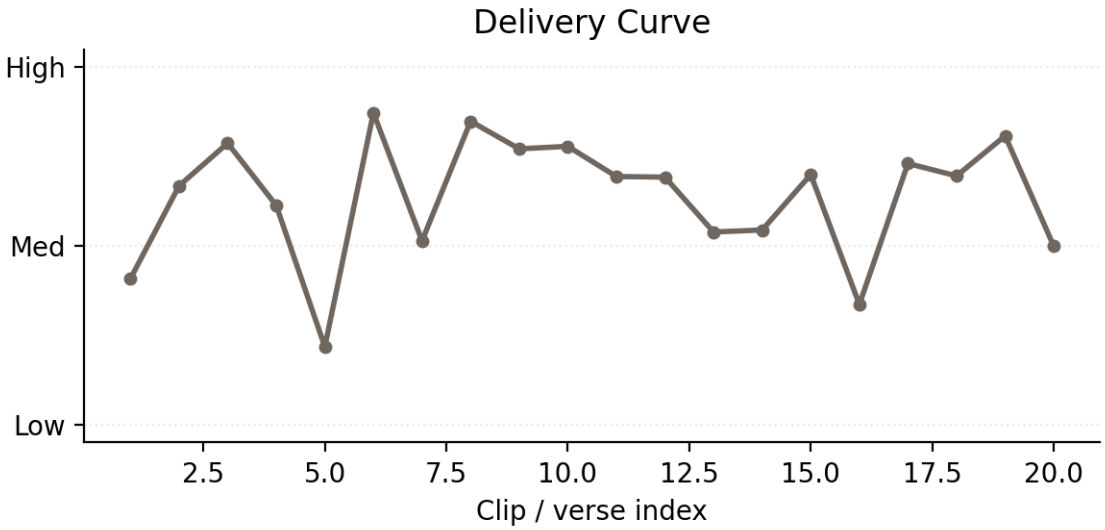

Figure 8: Arousal delivery curve — poem 0070, 20 clips. Arousal arc oscillates Medium/High with one Low dip, tracing the performer's emotional structure.

7. Key Insights

1 — Multitask sharing is the headline result. Genre and emotion taxonomies share latent structure (Hikma→Contemplation, Ghazal→Delicate Love). A single AraPoemBERT encoder trained jointly gives 4.1× genre F1 lift (0.132→0.543) at zero parameter cost. At this dataset scale, emotion supervision provides enough co-regularisation to prevent the genre head from overfitting to the Hikma-dominated split.

2 — ASR fine-tuning has a hard floor at ~3K clips. Seven LoRA runs covered every major adaptation axis; all regressed. R4 (frozen encoder) proves the forgetting is decoder-level. Actionable threshold: do not fine-tune Whisper on dialectal data below ~30K clips.

3 — Self-supervised contrastive learning beats supervised on heavily imbalanced audio. SimCLR NT-Xent (0.088 F1) outperforms supervised CNN-R3c (0.060 F1). Cross-entropy collapses every class to the majority on a 12-class imbalanced task; NT-Xent pushes all representations apart regardless of label frequency.

8. Limitations & Reproducibility

Limitations:

- Test set: 13 poems / 7 poets — CIs are wide; no poem-level significance claims.
- Hikma = 47% of test — macro-F1 is structurally suppressed.
- Corpus is 36 poets, predominantly male; may not generalise to female Nabati poets or other GCC regions.
- DMS rate 30.8% is irreducible without subjective "intended emotion" ground-truth.

Reproducibility:

Seeds: `torch.manual_seed(42)`, `np.random.seed(42)`, `random.seed(42)`. All `from_pretrained` calls use `local_files_only=True`. Hardware: Apple M-series (MPS auto-selected); CPU fallback on any platform. `just train` ~4–6 h; `just evaluate` ~5 min; 52 tests pass.

References

Radford et al. (2022). *Robust speech recognition via large-scale weak supervision.* arXiv:2212.04356. — Whisper. | Antoun et al. (2020). *AraBERT.* arXiv:2003.00104. | Farahani (2021). *AraPoem-BERT.* HuggingFace: `faisalq/bert-base-arapoembert`. | Johnson et al. (2019). *Billion-scale similarity search with GPUs.* IEEE Trans. Big Data. — FAISS. | Chen et al. (2020). *SimCLR.* ICML. | Holes & Abu Athera (2009). *Poetry and politics in contemporary Bedouin society.* Ithaca Press. | Al-Manea & Sowayan (2011). *Nabati poetry: The oral poetry of Arabia.*